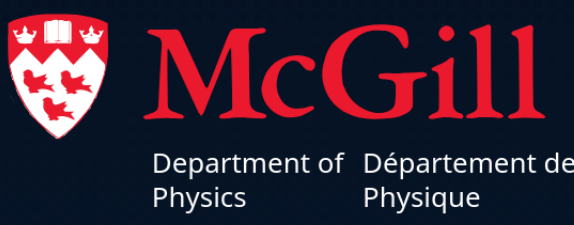


Sharing the Sky

Connecting Canadian FRB Discoveries to the Global Transient Community

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Alert fast. Stay reliable. How do you do both?

Diffuser vite. Rester fiable. Comment réaliser les deux ?

GLOSSARY

TRANSIENTS

Astrophysical events that vary in intensity over time — from milliseconds to years. Includes GRBs, supernovae, gravitational waves, and FRBs.

FRB

Fast Radio Bursts are intense, millisecond-duration bursts of radio waves of extragalactic origin.

CHIME

The Canadian Hydrogen Intensity Mapping Experiment — a radio telescope in Okanagan, BC. CHIME/FRB has detected over 4,000 FRBs.

GCN

NASA General Coordinates Network: the alert network used by the multi-messenger astronomy community for GW events, GRBs, neutrinos — and now FRBs.

VOEevent ALERT

A standardized XML format for machine-readable astronomical alerts. Observatories can automatically receive and act on transient detections.

TNS

IAU Transient Name Server: the official system for permanent, citable transient designations, indexed by NASA ADS.

CHIME/FRB & REAL-TIME ALERTS

CHIME/FRB detects about 600 Fast Radio Bursts per year. For a millisecond event, follow-up across wavelengths must be triggered immediately; due to the signal's dispersion, **we are already behind upon detection**.

Since 2021, CHIME/FRB has distributed real-time alerts through the **frb-voe** service at an average latency of about 12 seconds. These alerts have triggered follow-up observations before formal announcements, publications, or ATels — including Swift observations of the brilliant nearby FRB20250316A.

VOEvent alerts are issued prior to human verification. Events later determined not to be FRBs are retracted, with daily retraction notices issued every 24 hours. Approximately 6% of CHIME/FRB VOEvent detection alerts are retracted.

~600 FRBs/yr

~12 sec VOE latency

~6% retracted

RECENT IMPROVEMENTS

In February 2026, CHIME/FRB joined **NASA's GCN**, bringing FRB alerts into the same ecosystem used for gravitational waves and gamma-ray bursts. GCN notices reach the multi-messenger community with no additional setup, at a total latency of **one minute from detection**.

In parallel, the previously manual **TNS naming process was fully automated** within CHIME/FRB's internal database. Each FRB is now automatically named upon confirmation by two human observers as part of the standard detection pipeline.

This upgrade created a regularly updated stream of validated FRB events, and raised a new question: **should newly named FRBs be made immediately public**, turning TNS into a second alert broker? Or should the proprietary period be maintained until publication?

~1 min GCN latency

Automated TNS naming

What should a transient-detecting telescope prioritise when sharing a discovery?

⚡ Speed

Every second counts for follow-up. Miss the window, miss the science.



🔒 Reliability

The community builds science on these detections. We owe them accuracy.

OUR ANSWER

Two streams, two purposes.

A **VOEvent/GCN alert** is issued within seconds of detection, before human verification. It is designed to be fast and is understood by the community as preliminary.

A **TNS name** is assigned after confirmation by two human observers. Even then, an event may require further work: calibration errors may emerge, a burst may be associated with a known repeater, or properties may need revision.

⚡ Alert Broker

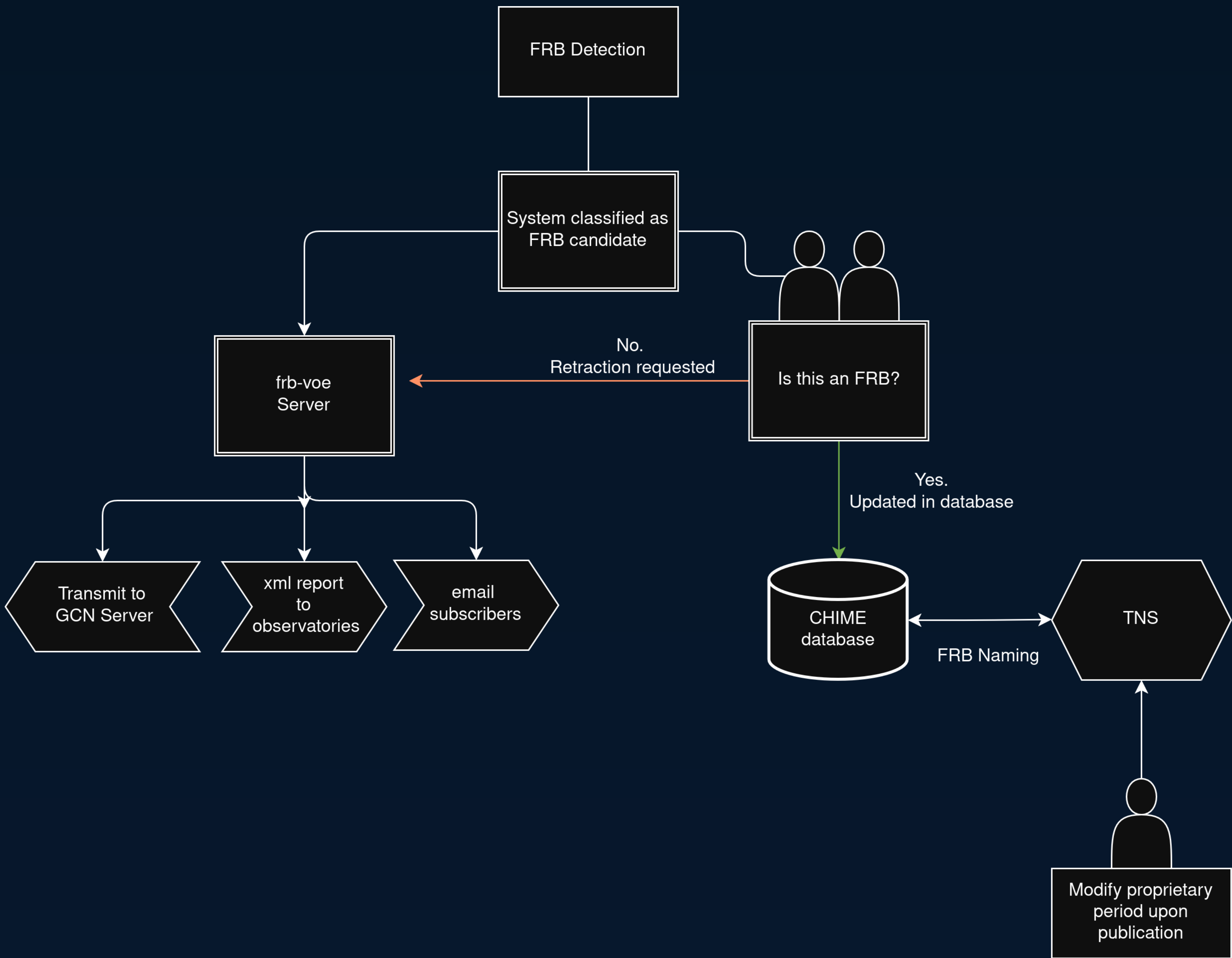
VOEvent / GCN. Seconds from detection. Fast, preliminary, designed to trigger follow-up.

🔒 Naming Catalog

TNS named after human verification, public upon publication. Curated and permanent.

REFERENCES

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CHIME/FRB detection pipeline – alert and naming streams